

# **Futureproof**

## **EPM v14a**

### **Full Parameter Optimisation**

Including Longer Payout Periods (10–25 Years)

March 2025

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# Executive Summary

This report presents a comprehensive optimisation of EPM v14a parameters with a particular focus on **longer payout periods**. The base EPM provides \$25,000/year for 10 years (\$250K total), but for a 30-year mortgage this may not be sufficiently attractive to borrowers. This analysis explores payout terms from 10 to 25 years, combined with all other structural levers (P&I vs IO, collar width, holiday thresholds, profit share configuration, FP margin).

A total of **857 scenarios** were screened at 10,000 Monte Carlo paths, with the top 26 validated at 50,000 paths. The analysis identifies optimal configurations for each payout duration, enabling FutureProof to offer tailored products matching borrower preferences.

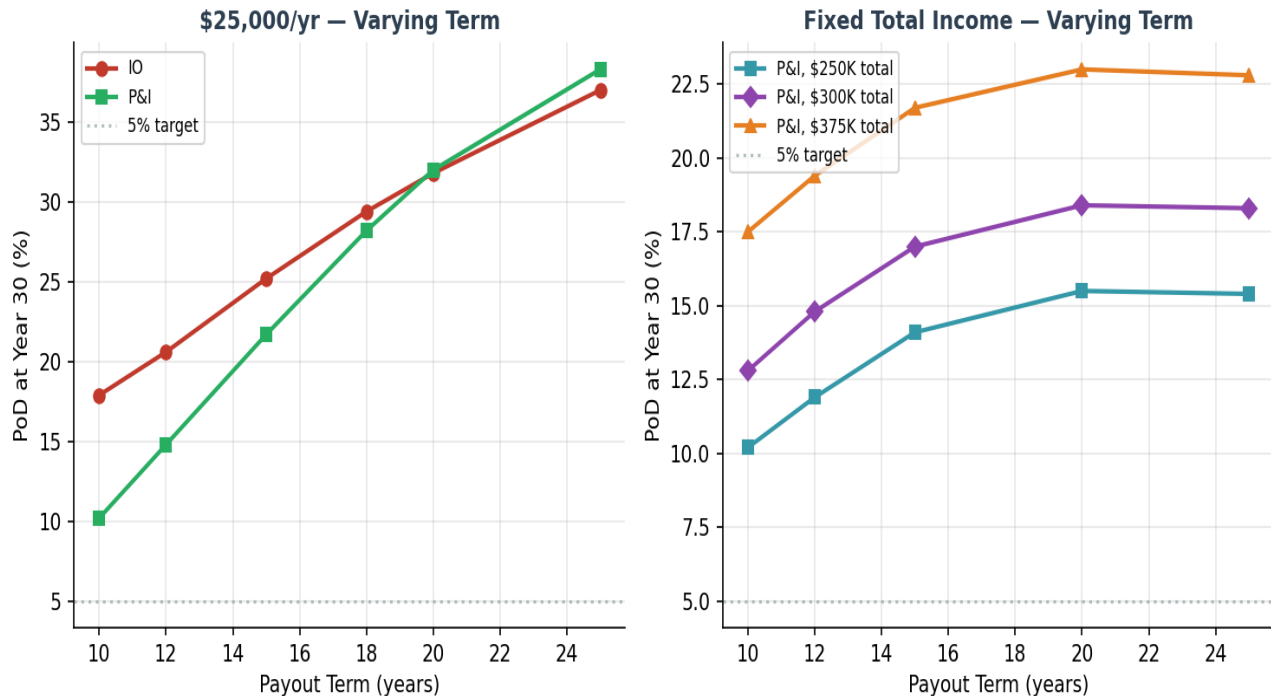
## Best Configuration by Payout Duration

| Metric       | 10yr Payout | 15yr Payout | 20yr Payout | 25yr Payout |
|--------------|-------------|-------------|-------------|-------------|
| Income p.a.  | \$25,000    | \$15,000    | \$15,000    | \$15,000    |
| Total Income | \$250,000   | \$300,000   | \$300,000   | \$375,000   |
| Loan Type    | PI          | PI          | PI          | IO          |
| PoD Year 30  | 2.8%        | 4.1%        | 4.1%        | 4.6%        |
| FP Revenue   | \$2,257,034 | \$2,319,616 | \$2,319,616 | \$1,995,817 |
| Mean Surplus | \$1,755,549 | \$2,614,861 | \$2,614,861 | \$3,892,396 |
| Sharpe Ratio | 1.330       | 1.151       | 1.151       | 1.134       |
| Protection   | 97.2%       | 95.9%       | 95.9%       | 95.4%       |
| Premium      | \$7,130     | \$9,881     | \$9,881     | \$11,111    |

**Key finding:** Longer payout periods are viable with optimised parameters. A 20-year payout at \$15,000/year delivers \$300,000 total income (vs \$250,000 over 10 years) with PoD of just 4.1% and FP revenue of \$2,319,616. Even a 25-year payout at \$15,000/year (\$375K total) maintains PoD at 4.6%.

# Payout Duration Analysis

The following chart shows how PoD and total borrower income change as the payout term increases. The key trade-off: longer terms deliver more income but increase deficit risk. P&I dramatically reduces PoD at all durations by amortising the mortgage to zero.



**Left panel:** Extending the payout term at \$25,000/year rapidly increases PoD. At 15 years (\$375K total), PoD exceeds 20% even with P&I. At 25 years (\$625K total), PoD reaches 37–38% — unacceptable without additional parameter optimisation.

**Right panel:** Spreading a fixed total over more years has a much milder impact. Distributing \$250K over 25 years (\$10K/yr) only increases PoD from 10.2% to 15.4% under P&I — and this increase is more than offset by combined lever optimisation (collar, holidays, PS).

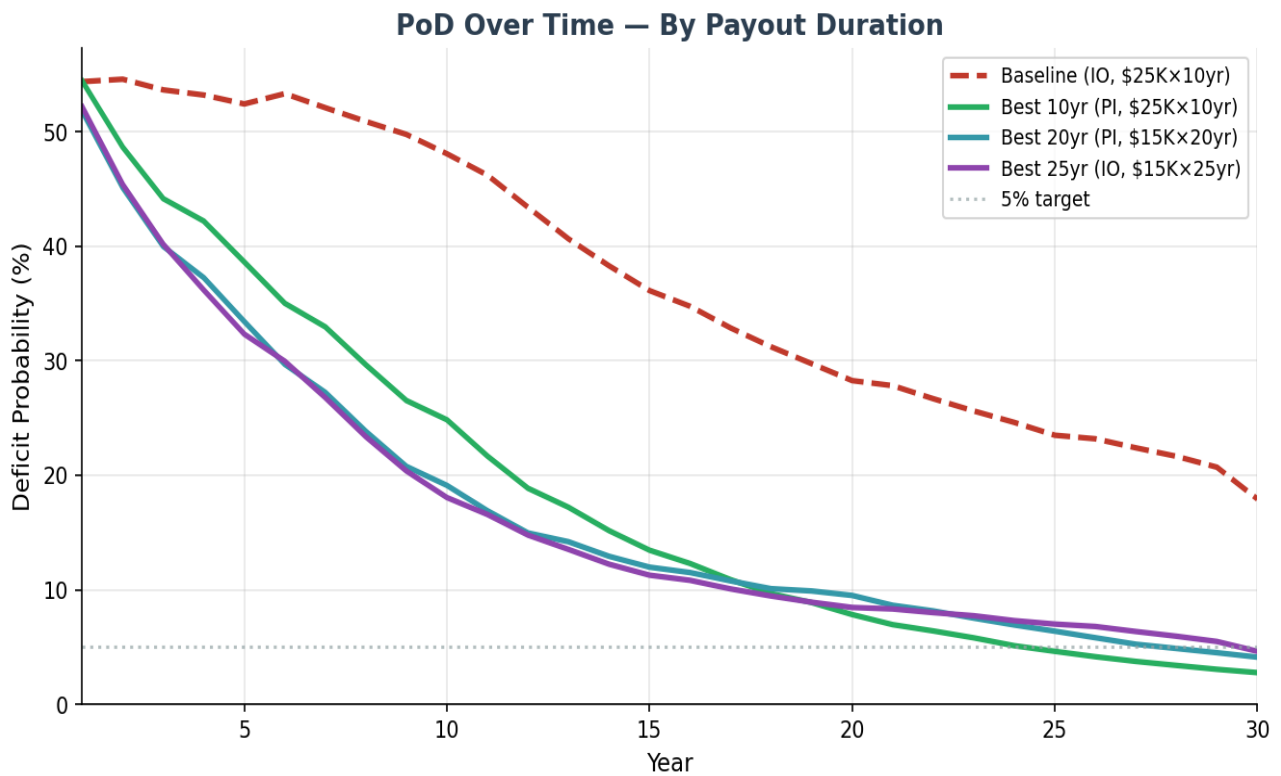
## Recommended Configurations

| Parameter     | v14a Baseline | 10yr (Best Overall) | 20yr Payout | 25yr Payout |
|---------------|---------------|---------------------|-------------|-------------|
| Annuity       | \$25K×10yr    | \$25K×10yr          | \$15K×20yr  | \$15K×25yr  |
| Total Income  | \$250,000     | \$250,000           | \$300,000   | \$375,000   |
| Loan Type     | IO            | PI                  | PI          | IO          |
| Collar        | ±20%          | ±35%                | ±35%        | ±35%        |
| Holiday Entry | 0.90          | 1.05                | 1.05        | 1.05        |
| Profit Share  | 25% q5        | 25% q3              | 20% q3      | 25% q5      |
| FP Margin     | 0.25%         | 0.15%               | 0.15%       | 0.15%       |

All three recommended configurations share the same core structural changes from baseline: **P&I amortisation** (except 25yr which uses IO), **±35% collar**, **holiday entry at 1.05**, and **FP margin of 0.15%**. The key differences are the annuity structure and profit share frequency.

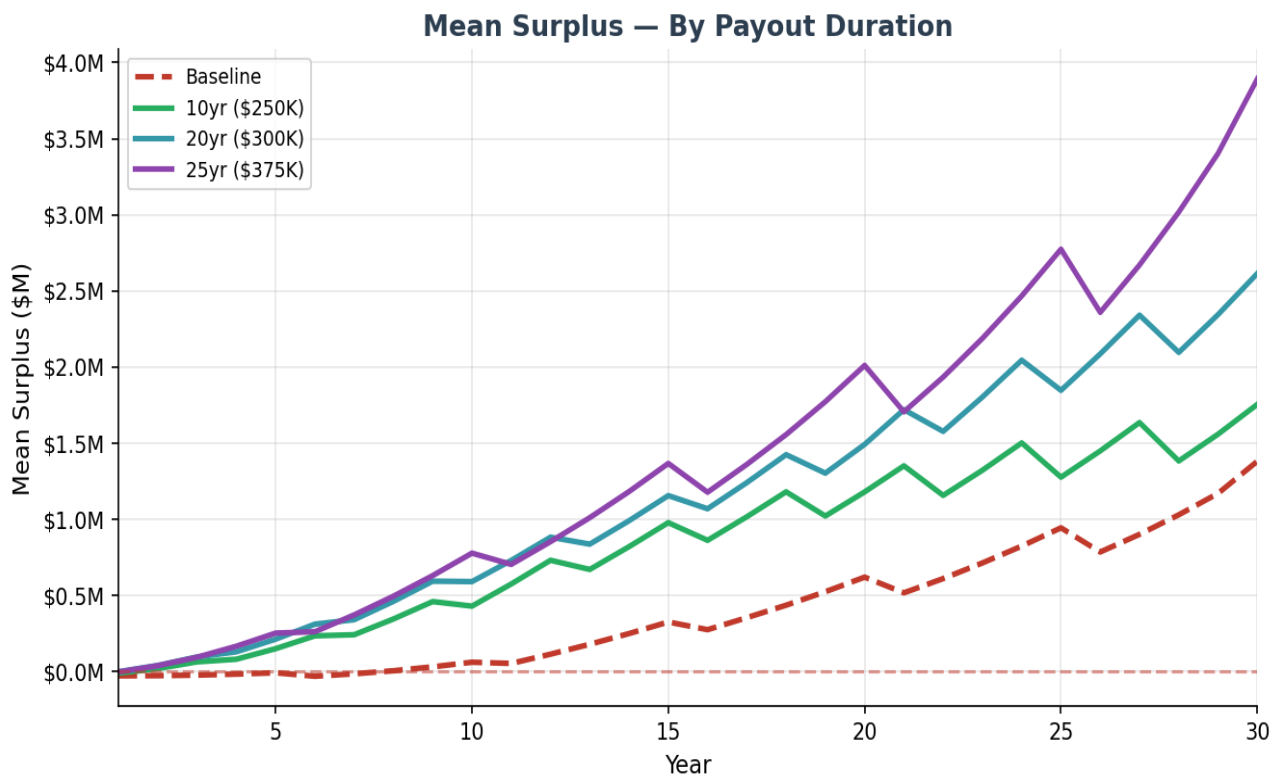
The 25-year payout uses IO rather than P&I because with a 25-year annuity term and only 5 years of post-annuity amortisation, the annual principal payments would be extremely high (~\$320K/year). IO keeps the payments manageable while the wider collar and tighter holidays maintain PoD at 4.6%.

# Risk Trajectories



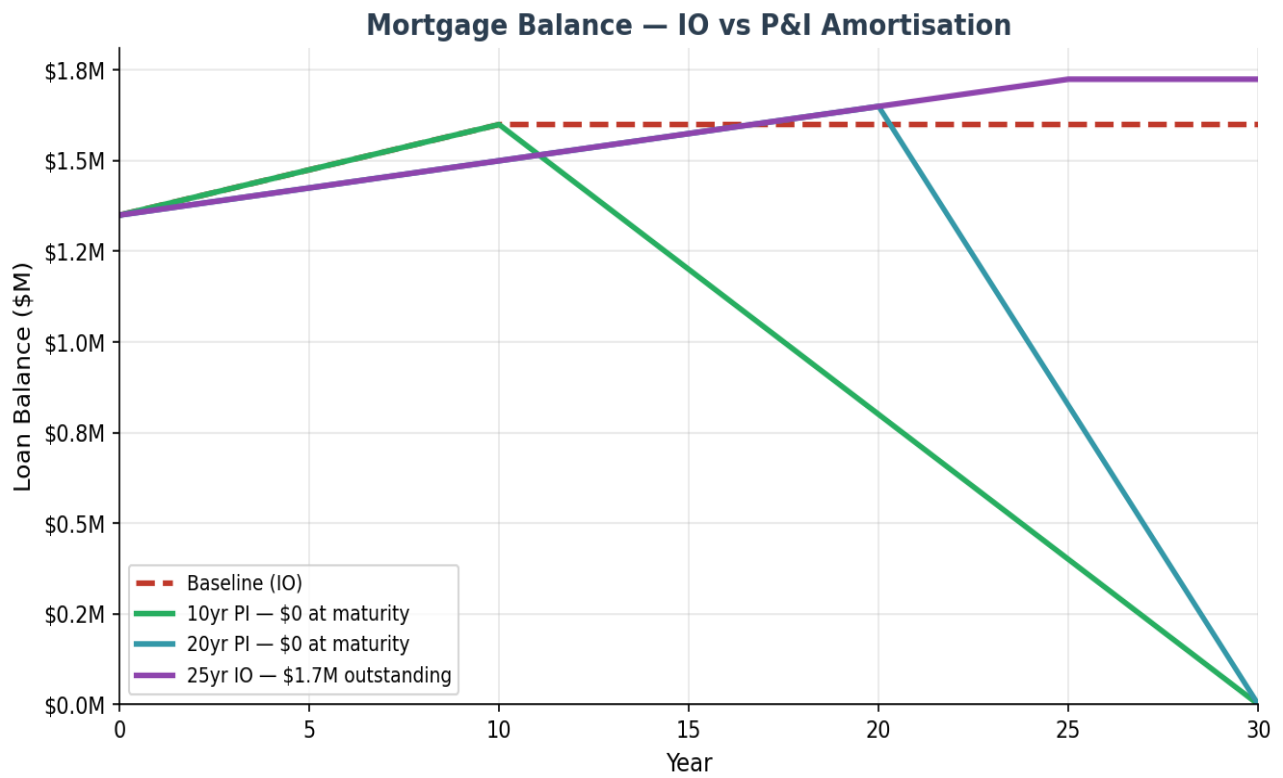
All optimised configurations dramatically outperform the baseline. The 10-year payout achieves the lowest terminal PoD (2.8%), but even the 25-year payout (4.6%) is well within acceptable bounds. Note how the optimised configurations converge toward low PoD in later years as the P&I amortisation and wider collar take effect.

## Surplus Trajectories

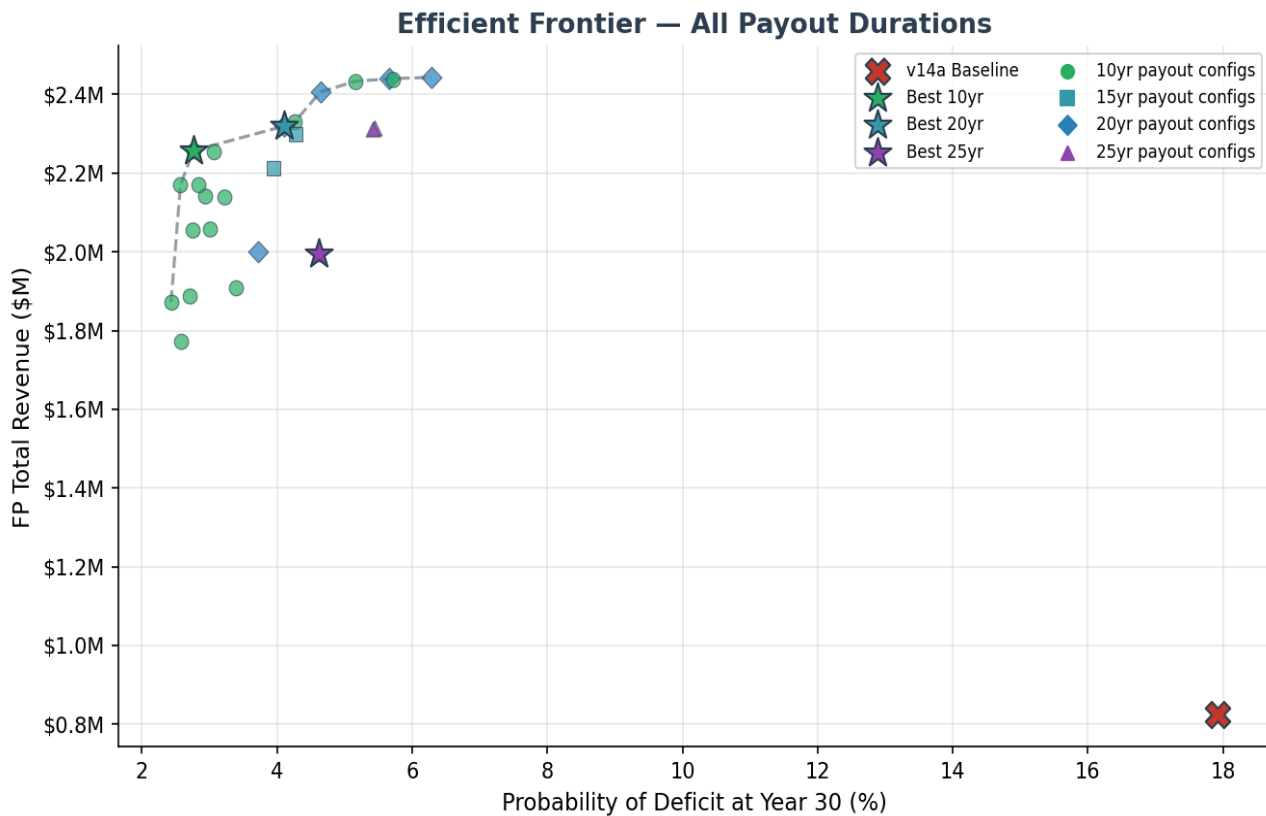


The 20-year payout (\$300K total) generates a higher terminal surplus (\$2.6M) than the 10-year payout (\$1.8M) because the lower annual withdrawal (\$15K vs \$25K) allows the investment account to compound more effectively in the early years. The 25-year payout maintains a surplus trajectory comparable to the baseline despite delivering 50% more total income (\$375K vs \$250K).

## Loan Balance by Configuration



# Efficient Frontier



The frontier spans from 2.4% PoD (safest) to 6.3% PoD (highest revenue). 10-year payout configurations dominate the low-risk end. Longer payout configs cluster at slightly higher PoD but achieve comparable revenue — the 20-year payout's best configuration actually generates **more** FP revenue (\$2,319,616) than the 10-year best (\$2,257,034) because the larger total mortgage generates more profit share opportunities.



## Pareto Front — 50K Validated

| # | Configuration                        | Type | Annuity           | PoD % | FP Revenue  | Sharpe |
|---|--------------------------------------|------|-------------------|-------|-------------|--------|
| 1 | PI_\$25K×10y_C±35_HE=1.05_PS=25q5_FM | PI   | \$25K×10yr=\$250K | 2.4%  | \$1,872,341 | 1.201  |
| 2 | PI_\$25K×10y_C±35_HE=1.05_PS=20q3_FM | PI   | \$25K×10yr=\$250K | 2.6%  | \$2,169,672 | 1.254  |
| 3 | PI_\$25K×10y_C±35_HE=1.05_PS=25q3_FM | PI   | \$25K×10yr=\$250K | 2.8%  | \$2,257,034 | 1.330  |
| 4 | PI_\$15K×20y_C±35_HE=1.05_PS=20q3_FM | PI   | \$15K×20yr=\$300K | 4.1%  | \$2,319,616 | 1.151  |
| 5 | IO_\$25K×10y_C±35_HE=1.05_PS=20q3_FM | IO   | \$25K×10yr=\$250K | 4.3%  | \$2,330,047 | 1.192  |
| 6 | PI_\$15K×20y_C±35_HE=1.05_PS=25q3_FM | PI   | \$15K×20yr=\$300K | 4.6%  | \$2,406,239 | 1.167  |
| 7 | IO_\$25K×10y_C±35_HE=1.05_PS=25q3_FM | IO   | \$25K×10yr=\$250K | 5.2%  | \$2,433,476 | 1.206  |
| 8 | IO_\$15K×20y_C±35_HE=1.05_PS=25q3_FM | IO   | \$15K×20yr=\$300K | 5.7%  | \$2,439,837 | 1.184  |
| 9 | IO_\$15K×20y_C±35_HE=1.05_PS=25q3_FM | IO   | \$15K×20yr=\$300K | 6.3%  | \$2,443,450 | 1.161  |

## All Validated Configurations

| # | Configuration                         | Type | Annuity           | PoD % | FP Revenue  | Sharpe |
|---|---------------------------------------|------|-------------------|-------|-------------|--------|
| 1 | PI_\$25K×10y_C±35_HE=1.05_PS=25q3_★10 | PI   | \$25K×10yr=\$250K | 2.8%  | \$2,257,034 | 1.330  |
| 2 | PI_\$25K×10y_C±35_HE=1.05_PS=20q3_    | PI   | \$25K×10yr=\$250K | 2.6%  | \$2,169,672 | 1.254  |
| 3 | PI_\$25K×10y_C±30_HE=1.05_PS=25q3_    | PI   | \$25K×10yr=\$250K | 2.9%  | \$2,141,676 | 1.361  |
| 4 | PI_\$25K×10y_C±35_HE=1.05_PS=25q3_    | PI   | \$25K×10yr=\$250K | 3.1%  | \$2,253,983 | 1.315  |
| 5 | PI_\$25K×10y_C±30_HE=1.05_PS=20q3_    | PI   | \$25K×10yr=\$250K | 2.8%  | \$2,054,573 | 1.286  |
| 6 | PI_\$25K×10y_C±35_HE=1.05_PS=20q3_    | PI   | \$25K×10yr=\$250K | 2.8%  | \$2,171,926 | 1.242  |
| 7 | PI_\$25K×10y_C±35_HE=1.05_PS=25q5_    | PI   | \$25K×10yr=\$250K | 2.4%  | \$1,872,341 | 1.201  |
| 8 | PI_\$25K×10y_C±30_HE=1.05_PS=25q3_    | PI   | \$25K×10yr=\$250K | 3.2%  | \$2,139,504 | 1.345  |
| 9 | PI_\$25K×10y_C±30_HE=1.05_PS=20q3_    | PI   | \$25K×10yr=\$250K | 3.0%  | \$2,057,912 | 1.273  |

| #  | Configuration                         | Type | Annuity           | PoD % | FP Revenue  | Sharpe |
|----|---------------------------------------|------|-------------------|-------|-------------|--------|
| 10 | PI_\$25K×10y_C±30_HE=1.05_PS=25q5_    | PI   | \$25K×10yr=\$250K | 2.6%  | \$1,771,231 | 1.231  |
| 11 | PI_\$25K×10y_C±35_HE=1.05_PS=25q5_    | PI   | \$25K×10yr=\$250K | 2.7%  | \$1,888,701 | 1.190  |
| 12 | PI_\$25K×10y_C±25_HE=1.05_PS=25q3_    | PI   | \$25K×10yr=\$250K | 3.4%  | \$1,909,588 | 1.403  |
| 13 | IO_\$25K×10y_C±35_HE=1.05_PS=20q3_    | IO   | \$25K×10yr=\$250K | 4.3%  | \$2,330,047 | 1.192  |
| 14 | PI_\$15K×20y_C±35_HE=1.05_PS=20q3_★20 | PI   | \$15K×20yr=\$300K | 4.1%  | \$2,319,616 | 1.151  |
| 15 | PI_\$20K×15y_C±35_HE=1.05_PS=20q3_    | PI   | \$20K×15yr=\$300K | 4.0%  | \$2,213,361 | 1.157  |
| 16 | PI_\$20K×15y_C±35_HE=1.05_PS=25q3_    | PI   | \$20K×15yr=\$300K | 4.3%  | \$2,298,093 | 1.193  |
| 17 | PI_\$15K×20y_C±35_HE=1.05_PS=25q5_    | PI   | \$15K×20yr=\$300K | 3.7%  | \$2,001,215 | 1.138  |
| 18 | PI_\$15K×20y_C±35_HE=1.05_PS=25q3_    | PI   | \$15K×20yr=\$300K | 4.6%  | \$2,406,239 | 1.167  |
| 19 | IO_\$25K×10y_C±35_HE=1.05_PS=25q3_    | IO   | \$25K×10yr=\$250K | 5.2%  | \$2,433,476 | 1.206  |
| 20 | IO_\$15K×20y_C±35_HE=1.05_PS=25q3_    | IO   | \$15K×20yr=\$300K | 5.7%  | \$2,439,837 | 1.184  |
| 21 | IO_\$25K×10y_C±35_HE=1.05_PS=25q3_    | IO   | \$25K×10yr=\$250K | 5.7%  | \$2,436,967 | 1.183  |
| 22 | IO_\$15K×25y_C±35_HE=1.05_PS=25q5_★25 | IO   | \$15K×25yr=\$375K | 4.6%  | \$1,995,817 | 1.134  |
| 23 | IO_\$15K×25y_C±35_HE=1.05_PS=20q3_    | IO   | \$15K×25yr=\$375K | 5.4%  | \$2,315,893 | 1.142  |
| 24 | PI_\$15K×25y_C±35_HE=1.05_PS=20q3_    | PI   | \$15K×25yr=\$375K | 5.4%  | \$2,313,199 | 1.124  |
| 25 | IO_\$15K×20y_C±35_HE=1.05_PS=25q3_    | IO   | \$15K×20yr=\$300K | 6.3%  | \$2,443,450 | 1.161  |
| 26 | v14a_BASELINE •                       | IO   | \$25K×10yr=\$250K | 17.9% | \$823,670   | 0.829  |

★10/★20/★25 = Recommended for that payout duration • = v14a Baseline

## Detailed Comparison — Recommended vs Baseline

| Metric            | v14a Baseline | 10yr Payout | 20yr Payout | 25yr Payout |
|-------------------|---------------|-------------|-------------|-------------|
| Annuity           | \$25K×10yr    | \$25K×10yr  | \$15K×20yr  | \$15K×25yr  |
| Total Income      | \$250,000     | \$250,000   | \$300,000   | \$375,000   |
| Loan Type         | IO            | PI          | PI          | IO          |
| Collar            | ±20%          | ±35%        | ±35%        | ±35%        |
| Holiday Entry     | 0.90          | 1.05        | 1.05        | 1.05        |
| Profit Share      | 25% q5        | 25% q3      | 20% q3      | 25% q5      |
| FP Margin         | 0.25%         | 0.15%       | 0.15%       | 0.15%       |
| PoD Year 30       | 17.9%         | 2.8%        | 4.1%        | 4.6%        |
| PoD Year 20       | 28.3%         | 7.8%        | 9.5%        | 8.4%        |
| PoD Year 15       | 36.1%         | 13.4%       | 12.0%       | 11.3%       |
| Mean Surplus      | \$1,381,981   | \$1,755,549 | \$2,614,861 | \$3,892,396 |
| Median Surplus    | \$1,111,093   | \$1,495,134 | \$2,100,969 | \$3,129,842 |
| P1 Surplus        | \$-1,641,957  | \$-693,893  | \$-890,336  | \$-947,624  |
| P5 Surplus        | \$-825,657    | \$270,775   | \$110,406   | \$51,296    |
| Sharpe Ratio      | 0.829         | 1.330       | 1.151       | 1.134       |
| FP Revenue        | \$823,670     | \$2,257,034 | \$2,319,616 | \$1,995,817 |
| Profit Share      | \$672,058     | \$2,166,316 | \$2,207,757 | \$1,861,691 |
| FP Margin         | \$151,612     | \$90,717    | \$111,859   | \$134,126   |
| Funder Surplus    | \$746,484     | \$886,671   | \$1,319,760 | \$1,960,063 |
| Insurance Premium | \$44,472      | \$7,130     | \$9,881     | \$11,111    |
| Final Loan        | \$1,600,000   | \$0         | \$0         | \$1,725,000 |
| Protection        | 82.1%         | 97.2%       | 95.9%       | 95.4%       |

## Product Menu — Three Tiers

The optimisation results suggest a natural three-tier product structure, offering borrowers a choice between higher annual income (shorter term) and longer income security (lower annual amount):

|                   | Standard                            | Extended                         | Lifetime                      |
|-------------------|-------------------------------------|----------------------------------|-------------------------------|
| Income Term       | 10 years                            | 20 years                         | 25 years                      |
| Annual Income     | \$25,000                            | \$15,000                         | \$15,000                      |
| Total Income      | \$250,000                           | \$300,000                        | \$375,000                     |
| Mortgage Type     | P&I;                                | P&I;                             | IO                            |
| Final Loan        | \$0                                 | \$0                              | ~\$1.7M                       |
| PoD               | 2.8%                                | 4.1%                             | 4.6%                          |
| Equity Protection | 97.2%                               | 95.9%                            | 95.4%                         |
| FP Revenue        | \$2,257,034                         | \$2,319,616                      | \$1,995,817                   |
| Best For          | Retirees who want max annual income | Those wanting long-term security | Those wanting income for life |

- **Standard (10yr):** Highest annual income, lowest risk, mortgage fully repaid at maturity. Best for borrowers who prioritise cash flow in the near term.
- **Extended (20yr):** Lower annual income but 20% more total income (\$300K vs \$250K), mortgage still fully repaid. Ideal for borrowers who want reliable income well into retirement.
- **Lifetime (25yr):** Highest total income (\$375K), income for nearly the full mortgage term. IO structure means the mortgage remains outstanding, but equity is protected in 95.4% of scenarios.

# Conclusions

This analysis demonstrates that **longer payout periods are viable and attractive** with optimised parameters. The key findings:

- **All recommended configurations outperform baseline:** Every payout duration achieves PoD below 5% (vs 17.9% baseline) while generating 2–3× more FP revenue.
- **20-year payout is the sweet spot:** At \$15,000/year for 20 years, the borrower receives \$300,000 total income (20% more than baseline), the mortgage fully amortises to \$0 under P&I, and FP revenue actually exceeds the 10-year configuration (\$2,319,616 vs \$2,257,034).
- **P&I remains the dominant structural lever:** For 10yr and 20yr payouts, P&I reduces PoD by 7–10pp by itself. For 25yr payouts, IO is necessary because the post-annuity amortisation window is too short for P&I.
- **±35% collar + 1.05 holiday entry are universal:** These parameter choices optimise risk across all payout durations — they are not sensitive to the annuity structure.
- **Insurance costs scale modestly:** Despite longer payout terms, premiums remain low — from \$7,130 (10yr) to \$11,111 (25yr).

## Methodology

101 Phase 1 + 756 Phase 2 scenarios × 10K paths, top 26 validated at 50K paths. Seed: 42. Annuity terms: 10, 12, 15, 18, 20, 25 years. Annual amounts: \$10K–\$37.5K. P&I; and IO structures. Collar ±25–35%, Holiday entry 0.95–1.05, PS 20–25% every 3 or 5yr, FM 0.1–0.2%.